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Assignment 202

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IARCO RESEACH PROPOSAL

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Research Topic	The equitable placement of additional vertical-evacuation refuges to improve tsunami safety in coastal Japan.

Equity Aware Prioritisation of Vertical-Evacuation Refuges for Tsunami Preparedness in Coastal Japan

Research Problem

A tsunami plan is effective only if residents can reach a safe location before inundation. In low lying coastal communities, high ground may be too distant, while existing evacuation buildings may be unreachable, unevenly distributed, or full. The planning problem is therefore where a limited number of additional vertical refuges would reduce the greatest time-to-safety gap. Japan's national evacuation guidance, facility resources, and Hazard Map Portal [1][2][3] provide an appropriate planning basis. However, a facility count cannot show whether refuges are reachable in time or whether they reduce unequal access between neighbourhoods.

This study aims to test the designation and capacity activation of a limited number of prescreened multistory buildings as vertical refuges. It compares an equity-aware investment rule with a proximity based rule and produces a shortlist for professional engineering, legal, and operational review.

Existing Literature

Distance alone cannot determine tsunami safety. Wood et al. showed that arrival time, walking speed, and earthquake related route failure can alter whether a pedestrian can reach safety [7]. Fraser et al. likewise found that departure delay and population exposure affect evacuation feasibility [8]. These studies support a time based access measure. Road constraints and refuge capacity are also highly material. Henry, Wood, and Frazier found that network bottlenecks can substantially change clearance time, while Nakai, Itatani, and Horiike showed that limited shelter capacity can leave evacuees unaccommodated [9][10]. A valid placement model must therefore include time allowance, route disruption, and capacity.

Few studies test how a fixed number of additional vertical refuge designations should be selected to reduce the worst neighbourhood level gap. This proposal addresses that intervention focused equity question.

Research Question and Hypotheses

Research Question: Given a fixed number of additional tsunami vertical evacuation designations, can an equity and robustness based site selection model reduce the maximum neighbourhood level safety deficit more effectively than a population proximity based selection model across stated tsunami evacuation scenarios in Japanese coastal municipalities?

The study will test three hypotheses:

1. Adding refuges selected by population proximity will improve average access but may leave a neighbourhood with a high share of residents who cannot reach safety in time.
2. Equity and robustness based selection will reduce the largest neighbourhood level safety deficit more than proximity based selection with the same number of interventions.
3. The equity aware strategy will require a tradeoff, reflected in greater mean walking time, a different site distribution, or lower aggregate coverage in some scenarios.

Methodology

I. Study Design and Intervention

The study will select four coastal municipalities that differ in terrain and settlement pattern. To be included, a municipality must have a current tsunami map, a refuge inventory, road and terrain data, population data, and a list of possible multi-storey buildings. The cases are intended to show how the method works in different settings, rather than to represent all Japanese coastal communities.

The intervention is the addition of vertical refuges from this pre-screened list. Before modelling, candidate sites will be limited to public or institution-owned multi-storey buildings that are shown in municipal records, can be mapped accurately, and are not already designated as tsunami refuges. Buildings will enter the model only as possible sites; their structural safety, usable evacuation level, ownership permission, and final capacity will require separate professional confirmation. The study will test the effect of adding one, three, and five sites in each municipality. The model will identify sites worth examining further; it will not decide whether a building should ultimately be designated or how much the project would cost.

II. Data Acquisition and Preprocessing

The study will use municipal hazard maps for inundation exposure and evacuation time allowances [3], GSI data for terrain [4], and National Land Numerical Information for roads [6]. The road lines will be converted into a walking network by linking verified intersections and noting any disconnected sections.

The smallest Census mesh available consistently across the four areas will be used as proxy demand [5]. Existing refuges and possible new sites will be taken from municipal lists. For each site, the study will record its location, designation, source date, capacity, and any missing information.

The evacuation time budget will be taken from the official municipal tsunami evacuation plan or from published tsunami-arrival-time information for the relevant hazard scenario. A municipality without an official planning-time allowance or arrival-time source will not be included in the final sample. Official refuge capacity will be used where it is available. Where capacity is not published, the site will be tested under low, medium, and high capacity assumptions rather than being assigned one unsupported value.

III. Calculation of Time-to-safety Margin

For each small population area i under each scenario s, the model will calculate a time-to-safety margin:

$$M_{i,s} = B_{i,s} - (D_s + R_{i,s})$$

Formula 1. Evacuation time budget balance equation

Symbol	Unit and Meaning
i	(No Unit) Index for the population mesh, or small origin area, being evaluated.
s	(No Unit) Index for the specified evacuation scenario being evaluated.
$M_{i,s}$	(Minutes) Time remaining after residents in area i have begun and completed the modelled evacuation in scenario s. 1. Positive - Time buffer

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	2. Zero - Arrival exactly at the stated deadline 3. Negative - insufficient modelled time
$B_{i,s}$	(Minutes) Evacuation time budget: the maximum no. of minutes available to leave area i and reach safety before the municipality specific tsunami planning deadline in scenario s.
D_s	(Minutes) Departure delay: time lost before evacuation begins in scenario s.
$R_{i,s}$	(Minutes) Route time: walking time from area i to the fastest refuge that is still reachable, safe, and has spare modelled capacity in scenario s.

Using this formula, if an area has an 18 minute evacuation time budget, a 5 minute departure delay, and a 10 minute feasible walking route, then $M_{i,s} = 18 - (5 + 10)$ leaves a 3 minute buffer. If no refuge can be reached with spare capacity, the mesh is recorded as no feasible refuge. However, it is crucial to note that this is a planning measure, not a survival prediction.

IV. Scenario Development

Scenarios will vary by the following four factors:

1. Early, moderate, and delayed departure
2. No additional disruption, limited disruption, & severe disruption of selected road links
3. Normal and reduced refuge capacity
4. Full availability or loss of one existing refuge

V. Site Selection Comparison

Three selection rules will be applied to the same candidate buildings and the same facility budget: no new refuges, refuges selected near the largest exposed proxy population, and refuges selected to reduce the highest neighbourhood safety deficit. Comparing the results indicates whether an equity aware choice of sites gives more neighbourhoods enough time to reach a feasible refuge than a simple proximity based choice, and whether this benefit requires longer average walking time or reduced overall coverage.

For each facility budget, a Python script will evaluate every feasible combination of candidate sites. It will retain the combination with the lowest maximum neighbourhood safety deficit; if two or more combinations produce the same deficit, it will select the combination with the lowest population-weighted total route time.

Strategy	Site Selection Rule	Purpose of Comparison
Baseline	No additional vertical refuges	Measures the existing modelled access gap
Proximity	Sites nearest the largest exposed proxy population	Represents an efficiency oriented investment rule
Equity Robustness	Sites that minimize the worst safety deficit then travel time	Tests the proposed intervention

VI. Outcome Measures and Analysis

A neighbourhood will be a predefined group of adjacent Census meshes. The primary outcome is the maximum neighbourhood safety deficit:

$$Neighborhood\ \ safety\ deficit\ (\%)\ =\ \frac{(proxy\ demand\ in\ a\ region\ with\ a\ negative\ time/to/safety\ margin)}{(total\ proxy\ demand\ in\ that\ neighborhood)} \times 100$$

This percentage will be calculated for every neighbourhood and report the highest value. Secondary outcomes are positive margin coverage, mean walking time, refuge utilisation, and unmet proxy demand.

Model Checking & Research Ethics

Before running the models, the researcher will use QGIS to overlay roads, hazard areas, Census meshes, and refuges, then use a Python routing script to check that every origin and refuge connects to the network. Duplicate links will be removed, disconnected segments will be recorded, and samples of calculated routes with the source maps will be compared manually. The script will then rerun the model with different walking speeds, delays, capacities, road disruptions, and neighbourhood groupings.No personal data will be used. Aggregate data will not be used to infer individual mobility, behaviour, or priority.

Limitations

The quality of the findings will depend on the available public data. Maps may not show informal routes, temporary closures, access restrictions, or recent population change. The model also cannot fully account for human behaviour, congestion, or earthquake damage. Any candidate building would still need professional confirmation of its structure, usable level, ownership, and capacity before it could be designated as a refuge.

Anticipated Post-Project Contribution

The completed study will show how the two investment rules affect the highest neighbourhood safety deficit, travel time, coverage, and refuge use under the same budget. It should also produce a preliminary list of sites that could be prioritised for structural review, capacity assessment, and access-agreement discussions.

Conclusion

This study asks whether the placement of a small number of additional vertical refuges can be made fairer without unnecessarily reducing overall access. By comparing the two selection rules, it will identify which candidate sites are most promising for further professional assessment.

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